

Process Flow — Full Lifecycle Reference

Nine diagrams tracing the full lifecycle — from an approved Oracle Purchase Requisition through contract creation, dynamic approval chains, progress payments, change orders, manpower reconciliation, SLA penalties, and performance scorecards.

Each diagram is annotated with the exact business rule it encodes.

Currency — SAR throughout Source — CLAUDE.md domain model & business rules

■ Start / terminal state

■ Decision point

■ Approval step

■ External system (Oracle / HCM)

■ Positive outcome

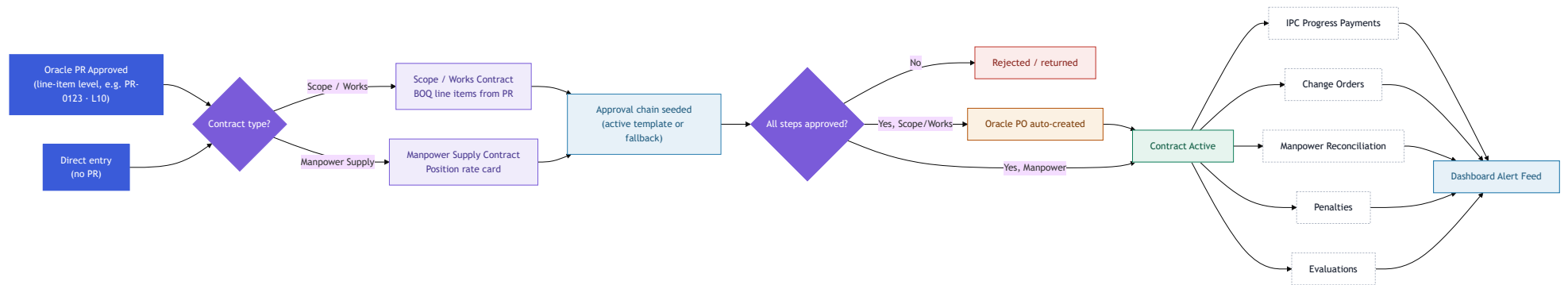
■ Negative outcome / stop

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01 End-to-end lifecycle

A contract always starts from one of two mutually exclusive origins, then flows into whichever of the five ongoing-operations tracks apply to it. All tracks ultimately surface into the dashboard alert feed.

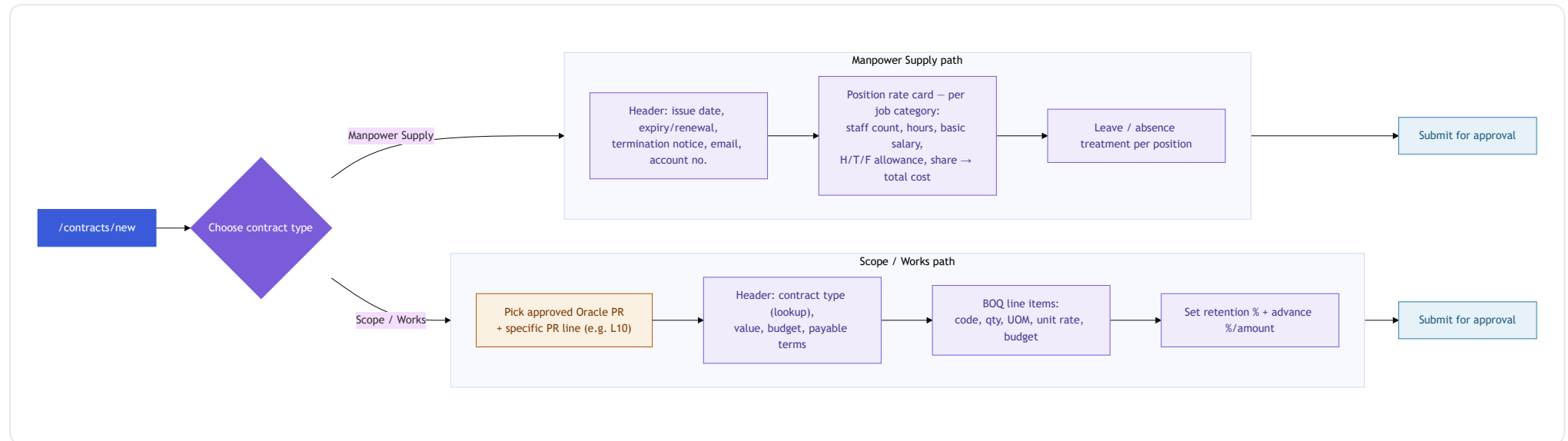


REFERENCE RULES

- Scope/Works and Manpower Supply are created via **completely separate flows that never share fields or validation**.
- Manpower Supply contracts **never** receive an Oracle PO — they are not PR/PO-triggered.

02 Contract creation — two independent paths

`/contracts/new` is a chooser only — the two forms downstream share no fields or validation logic.

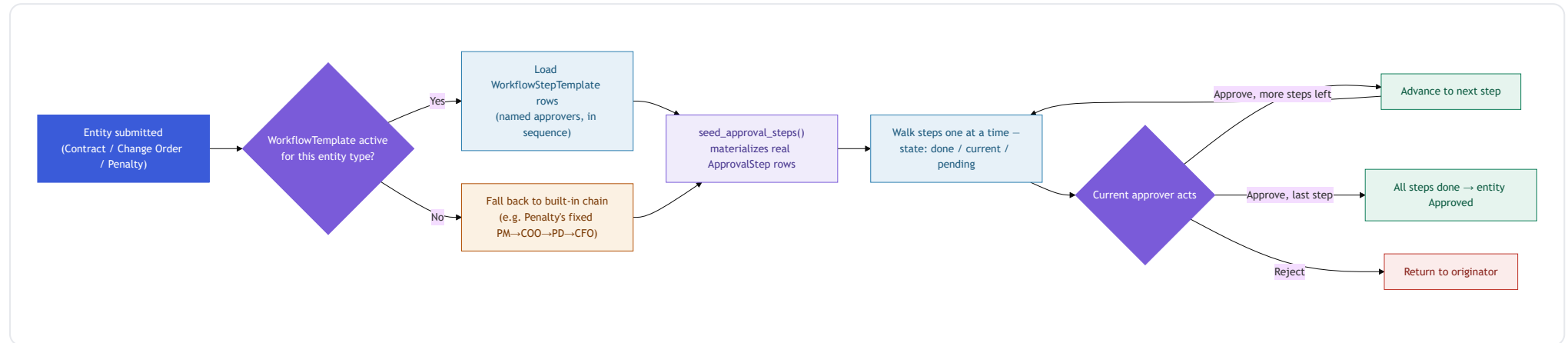


REFERENCE RULES

- Contract Type (e.g. "Hard FM (MEP)", "Manpower") is picked from the maintained **ServiceTypeOption** lookup, never free text.
- Manpower Supply has **no retention**, **no advance**, **no payable terms** — a rate card replaces the BOQ.

03 Dynamic approval flow engine

Every approvable entity (Contract, Change Order, Penalty) asks the same question at submit time: is there an active, admin-built chain for this entity type?

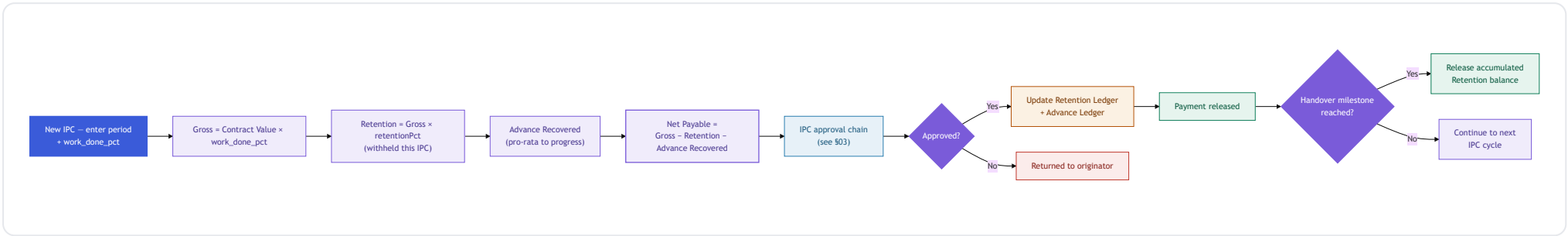


REFERENCE RULES

- Only **one flow per entity type** can be active at a time; flows are strictly linear — no branching or conditional approval.
- **AppUser** is a lightweight directory assignable to steps — there is no login/session/current-user concept anywhere in the app.

04 IPC progress payment cycle (Scope/Works only)

Every Interim Payment Certificate reduces to one formula, then walks the same approval engine as §03 before releasing cash.

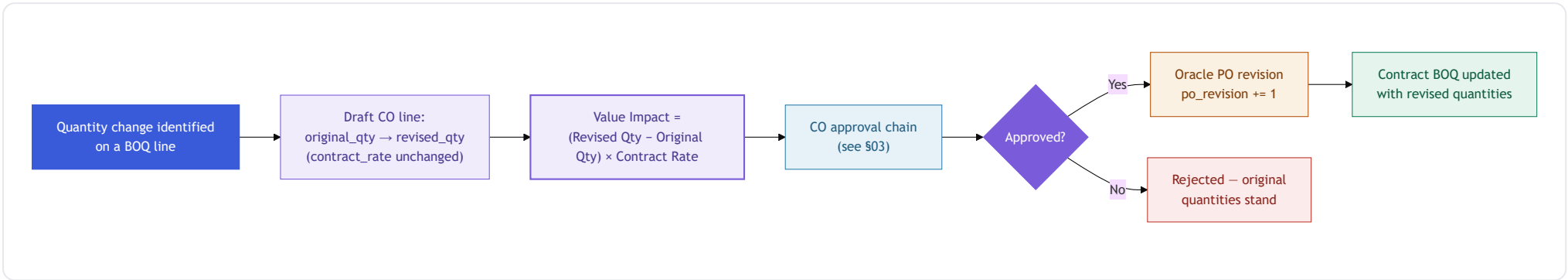


REFERENCE RULES

- Retention is withheld per IPC at **retentionPct** and released only on handover.
- Advance is paid on mobilisation and recovered **pro-rata to progress** across subsequent IPCs.

05 Change order flow (Scope/Works only)

Change Orders vary **quantities only** — unit rates are locked to the original contract baseline for the life of the contract.

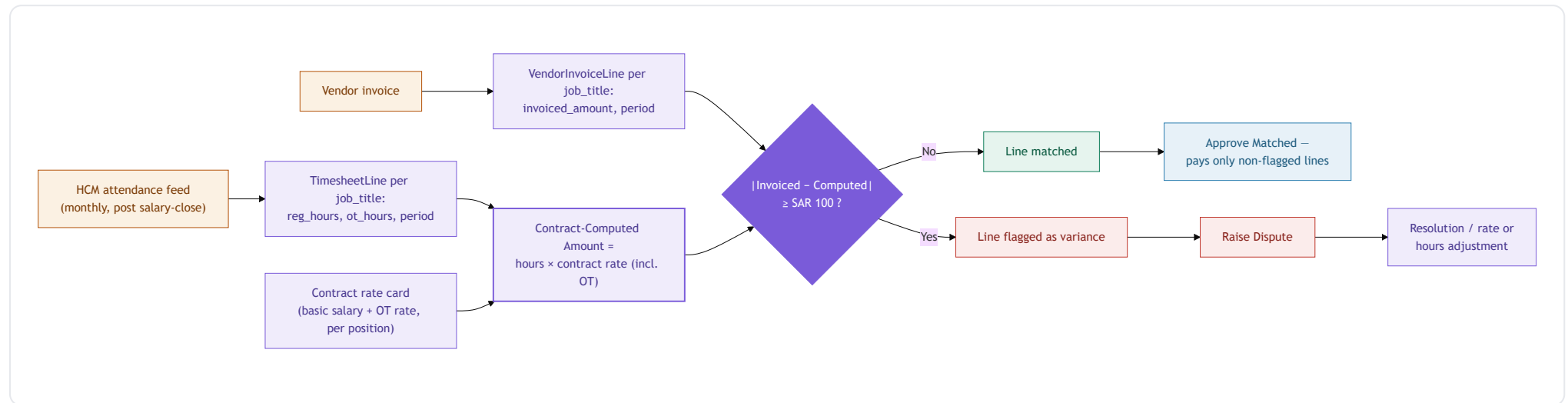


REFERENCE RULES

- An approved Change Order creates a **PO revision** (increments the existing PO's revision number) — never a new PO.

06 Manpower reconciliation (Manpower Supply only)

Monthly HCM attendance is compared line-by-line against the vendor invoice, using the contract rate card — including OT rates — as the baseline of truth.

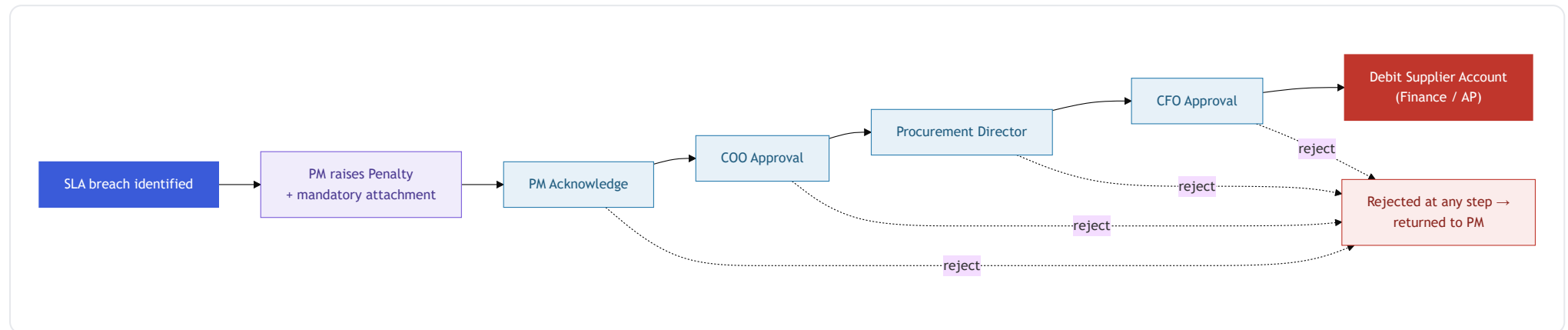


REFERENCE RULES

- Variance threshold is a fixed **SAR 100** absolute difference per job title, per period.
- "Approve Matched" pays only the non-flagged lines — flagged lines are held out for dispute, never auto-paid.

07 Penalty approval chain

The one flow in the system with a fixed, non-configurable-by-default fallback sequence — five roles, strictly in order, no skipping.

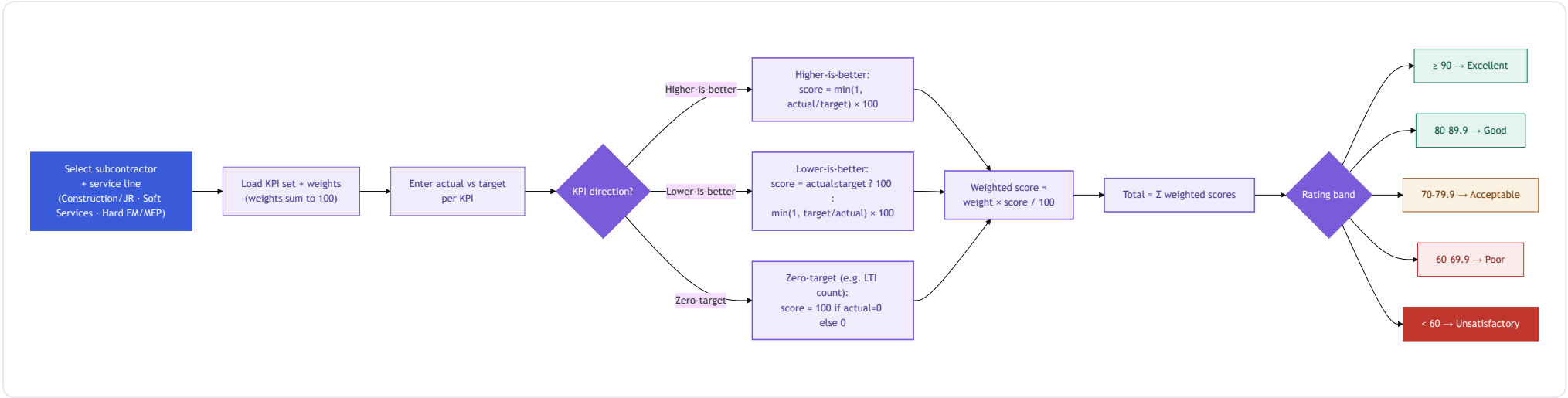


REFERENCE RULES

- Attachment is **mandatory** before a penalty can be raised — no exceptions.
- The supplier account is debited **only after CFO approval** — never earlier in the chain.
- If an active **WorkflowTemplate** exists for Penalty, it replaces this fixed chain (see §03) — this diagram is the fallback.

08 Subcontractor evaluation scorecard

Each KPI scores independently by its direction, then rolls up through its weight into a single rating band.

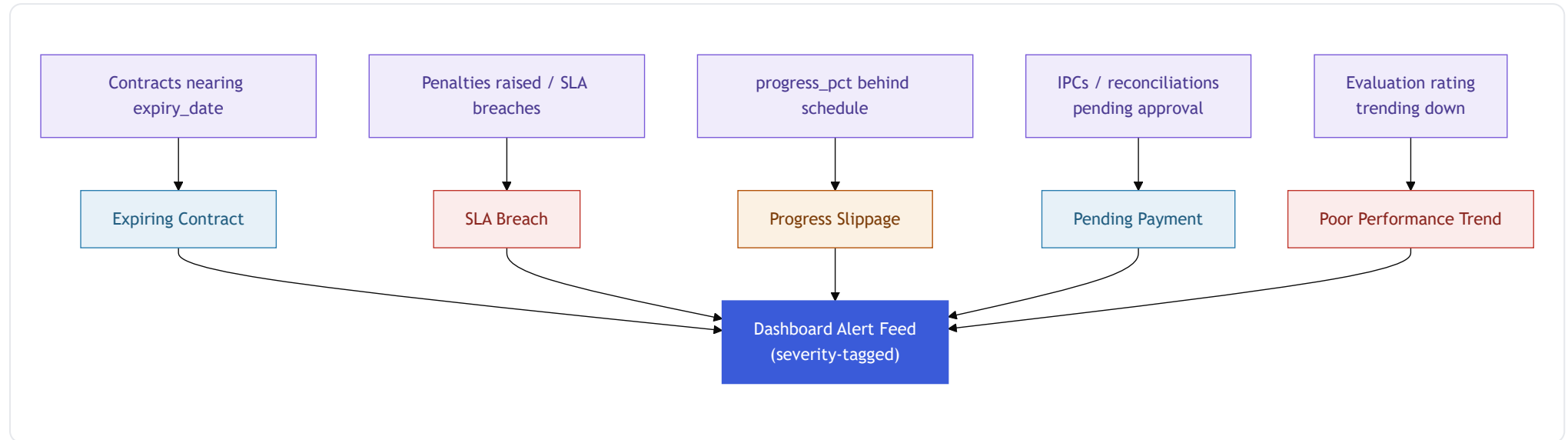


REFERENCE RULES

- All three scoring rules cap at **100** — over-performance never pushes a KPI score above 100.

09 Dashboard alert generation

Alerts are never hand-authored — they are the output of five standing queries run across every live contract.



REFERENCE RULES

- Alerts are **auto-generated** from live query results — there is no manual "create alert" action anywhere in the app.